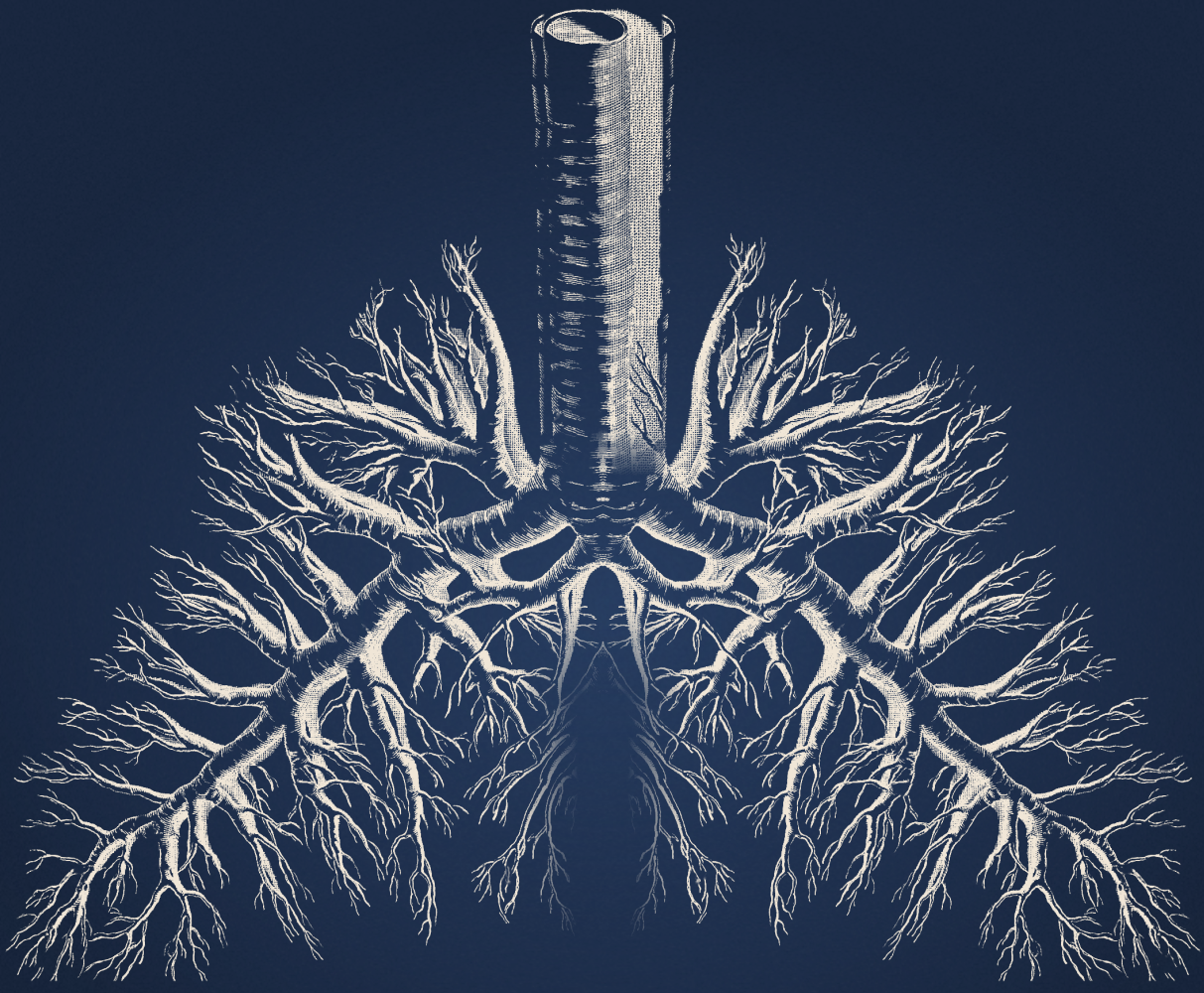
ADULT CRITICAL CARE

THE ICU EQUATIONS

300 formulas, calculations and scores



MOHSIN ALMANAMI

ADULT CRITICAL CARE

The ICU Equations

300 formulas, calculations and scores for adult critical care

Mohsin Almanami

The ICU Equations
300 formulas, calculations and scores for adult critical care
First edition, 2026

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Cover illustration adapted from an engraving in the Wellcome Collection, public domain: the trachea and bronchi, after Gérard de Lairesse (1739).

Medical disclaimer. This book is a reference for qualified healthcare professionals. Formulas, thresholds and doses were checked against the cited sources at the time of writing (October 2026). Practice changes and local protocols differ: confirm drug doses against the current product label and your institution's guidelines before use. The book supports, and does not replace, clinical judgment. The author accepts no liability for decisions made on the basis of its content.

Values are for adults unless stated otherwise. Units follow the cited source; conversions are in Appendix A.

Set in Playfair Display, Source Serif 4 and Manrope. Equations typeset with KaTeX.

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BEFORE YOU START

How to use this book

Each formula has its own page, in the same order and the same positions throughout, so the eye learns where to look.

Running head and tab. Part number and name at the top; the colored tab on the fore-edge marks the part, so a part can be found with the book closed. Each part keeps one color from its opening page to its last formula.

Formula ID and level. IDs (for example VENT-009) are used for cross-references. **Core**: applied daily at the bedside by every ICU clinician. **Advanced**: used in specific situations or advanced assessment. **Specialized**: narrow clinical or research use.

The equation sits under the colored rule, followed by its variables and units. Units are the most frequent source of calculation error: check them before you calculate.

Normal, target and alert values. The shaded panel gives the reference range and, in **red**, the threshold that should prompt action where a validated threshold exists. Red is used for nothing else in the book; where no numeric threshold exists the alert text is set in black.

Worked example in the dark panel, using realistic bedside values.

Interpretation, common mistake and limitation sit at the foot of the entry, above the source; when an entry needs a second page, they continue on the following page.

Source. The primary reference for every formula is printed at the foot of its page. Where the reference has a stable online record, the QR code opens it (PubMed or DOI). The full list, with links, is in the References section. **See also** gives related formulas with their page numbers.

PART

01

Acid–base & blood gases

25 formulas · ABG-001 to ABG-025

ABG-001	Henderson-Hasselbalch Equation
ABG-002	Henderson Equation (Simplified)
ABG-007	Winter's Formula (Metabolic Acidosis Compensation)
ABG-008	Acute Respiratory Acidosis Compensation
ABG-009	Chronic Respiratory Acidosis Compensation
ABG-010	Acute Respiratory Alkalosis Compensation
ABG-011	Chronic Respiratory Alkalosis Compensation
ABG-012	Metabolic Alkalosis Compensation
ABG-003	Anion Gap (AG)
ABG-004	Albumin-Corrected Anion Gap
ABG-005	Delta Gap
ABG-006	Delta Ratio (Delta/Delta)
ABG-024	Urine Anion Gap
ABG-013	Base Excess / Standard Base Excess
ABG-014	Buffer Base
ABG-015	Strong Ion Difference (SIDa)
ABG-016	Strong Ion Gap (SIG)
ABG-017	Calculated Serum Osmolality
ABG-018	Osmolar Gap
ABG-019	Corrected Sodium for Hyperglycemia
ABG-020	Corrected Calcium for Albumin
ABG-021	Free Water Deficit
ABG-022	Adrogue-Madias Formula
ABG-023	Bicarbonate Deficit
ABG-025	Temperature Correction of ABG (alpha-stat vs pH-stat)

PART 01 · OVERVIEW

Acid–base & blood gases

The arterial blood gas is read in three steps. The first establishes the pH, the second identifies the process driving it, and the third checks whether the body's response to that process is the one expected. The ICU measures acid–base state because respiratory failure, renal dysfunction, shock and poisoning often coexist, and a single pH value can conceal opposing disturbances. The part also covers the osmolality, electrolyte and temperature calculations requested alongside the gas.

Internal consistency of a reported gas is tested first, using the Henderson-Hasselbalch equation (ABG-001) and its simplified form (ABG-002). The compensation rules follow: Winter's formula (ABG-007), the acute and chronic respiratory acidosis rules (ABG-008, ABG-009), the acute and chronic respiratory alkalosis rules (ABG-010, ABG-011) and the metabolic alkalosis rule (ABG-012). Each predicts a value that is compared with the measurement. The anion gap (ABG-003) then classifies metabolic acidosis, and its albumin correction (ABG-004), the delta gap (ABG-005) and the delta ratio (ABG-006) look for a second metabolic disorder. The urine anion gap (ABG-024) addresses normal-gap acidosis. Base excess (ABG-013), buffer base (ABG-014), the strong ion difference (ABG-015) and the strong ion gap (ABG-016) offer whole-blood and physicochemical views. Osmolality (ABG-017) and the osmolar gap (ABG-018) lead into the corrections and deficits: sodium (ABG-019), calcium (ABG-020), free water (ABG-021), the Adrogue-Madias estimate (ABG-022) and bicarbonate (ABG-023). Temperature correction (ABG-025) closes the part.

Sample and context govern every result. Use an arterial sample, pair it with electrolytes drawn at the same time, and confirm units before calculating, because a $PaCO_2$ in kPa entered as mmHg corrupts every later step. The compensation formulas are population averages and each is valid only for the disorder it describes. Albumin, phosphate and paraproteins alter the anion gap, so its interpretation depends on the patient's protein status. Analyzers report at 37 °C, and the effect of correcting for body temperature is debated. Most corrections here are approximations.

ABG-001 · CORE FORMULA

Henderson-Hasselbalch Equation

01

Quantifies pH from HCO_3^- and $PaCO_2$, linking acidity to its metabolic and respiratory components and underpinning arterial blood gas (ABG) interpretation.

$$pH = 6.1 + \log_{10}\left(\frac{HCO_3^-}{0.03 \times PaCO_2}\right)$$

HCO_3^- plasma bicarbonate (mmol/L) $PaCO_2$ arterial CO_2 tension (mmHg)

NORMAL / TARGET	pH 7.35–7.45; $PaCO_2$ 35–45 mmHg; HCO_3^- 22–26 mmol/L
ALERT	pH < 7.20 or > 7.55: imminent danger requiring immediate intervention
USED BY · LEVEL	Physician · Respiratory Therapist · Nurse · Core

WORKED EXAMPLE

Inputs	HCO_3^- 24 mmol/L; $PaCO_2$ 40 mmHg	
Step 1	0.03 × 40 = 1.2 mmol/L	
Step 2	24 ÷ 1.2 = 20	
Step 3	$\log_{10}$ 20 = 1.301	pH = 7.40
Step 4	6.1 + 1.301 = 7.40	normal pH

INTERPRETATION	A 20:1 ratio of HCO_3^- to dissolved CO_2 yields a normal pH. Any deviation from this ratio shifts pH.
COMMON MISTAKE	Using $PaCO_2$ in kPa without converting (kPa × 7.5 = mmHg), omitting the 0.03 coefficient and mixing measured with calculated HCO_3^- each produce errors.
LIMITATION	The equation does not describe complex multi-cause disorders. The pKa shifts with severe hypoalbuminemia and hypothermia.

SCAN PubMed

Source Marino PL. Marino's The ICU Book. 4th ed. Philadelphia: Wolters Kluwer/Lippincott Williams & Wilkins; 2014. | Kellum JA. Crit Care 2005;9(5):500-507

See also Anion Gap **p. 15**

ABG-002 · ADVANCED FORMULA

Henderson Equation (Simplified)

01

Checks the internal consistency of an arterial blood gas (ABG), showing whether the result is mathematically plausible or a laboratory error.

$$[H^+]\,(\text{nmol/L}) = 24 \times \frac{PaCO_2}{HCO_3^-}$$

$[H^+]$	hydrogen ion concentration (nmol/L)	$PaCO_2$	arterial CO_2 tension (mmHg)
HCO_3^-	bicarbonate (mmol/L)		

NORMAL / TARGET	$[H^+]$ 36–44 nmol/L (pH 7.35–7.45)
ALERT	$[H^+]$ > 60 nmol/L (pH < 7.22) or < 26 nmol/L (pH > 7.58)
USED BY · LEVEL	Physician · Respiratory Therapist · Advanced

WORKED EXAMPLE

Inputs	$PaCO_2$ 60 mmHg; HCO_3^- 24 mmol/L	$[H^+]$ = 60 nmol/L, pH ≈ 7.22
Step	24 × (60 ÷ 24) = 60 nmol/L	acidemia

INTERPRETATION	Each 1 nmol/L rise in $[H^+]$ above 40 lowers pH by ≈ 0.01.
COMMON MISTAKE	Forgetting that the relationship is logarithmic makes estimates inaccurate outside pH 7.20–7.50.
LIMITATION	The formula is an approximation. Accuracy degrades outside ≈ pH 7.20–7.50.

SCAN SOURCE

Source Kellum JA, Elbers PWG, editors. Stewart's Textbook of Acid-Base. 2nd ed. Amsterdam: Lulu Enterprises; 2009.

See also Henderson-Hasselbalch equation **p. 7**

VENT-003 · CORE FORMULA

Static Compliance (Cstat)

03

Quantifies the elastic recoil of the total respiratory system (lung plus chest wall) under no-flow conditions, a core severity marker that informs ventilator strategy.

$$C_{stat} = \frac{V_T}{P_{plat} - PEEP_{total}}$$

V_T tidal volume (mL)

P_{plat} plateau pressure (cmH_2O)

$PEEP_{total}$ set + auto-PEEP (cmH_2O)

NORMAL / TARGET	60–100 mL/cmH_2O (normal, passive ventilation)
ALERT	< 30–40 mL/cmH_2O: suggests severe restrictive lung disease or ARDS
USED BY · LEVEL	Respiratory Therapist · Physician · Core

WORKED EXAMPLE

Inputs V_T 420 mL; P_{plat} 28 cmH_2O; PEEP 8 cmH_2O

Step 1 28 − 8 = 20 cmH_2O

Step 2 420 ÷ 20 = 21 mL/cmH_2O

Cstat = 21 mL/cmH_2O severely reduced compliance

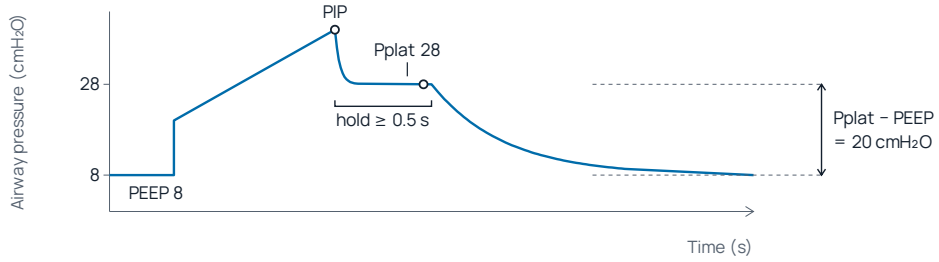


Figure 3.1 Airway pressure during a volume-controlled breath with an end-inspiratory hold: P_{plat} is read on the plateau, and Cstat is V_T divided by the span from PEEP to P_{plat}.

INTERPRETATION	A compliance this low indicates a very stiff respiratory system, consistent with severe ARDS.
COMMON MISTAKE	Measuring during active patient effort (breath stacking or spontaneous inspiration) instead of a true passive plateau gives an invalid value.
LIMITATION	Requires a brief inspiratory pause of 0.5–2 s with no patient effort. Invalid with active breathing or leaks.

SCAN SOURCE

Source Marino PL. Marino's The ICU Book. 4th ed. Philadelphia: Wolters Kluwer/Lippincott Williams & Wilkins; 2014.

See also Dynamic compliance **p. 64** · Driving pressure **p. 75**

VENT-004 · CORE FORMULA

Dynamic Compliance (Cdyn)

03

Reflects combined elastic and resistive load during active flow; used with static compliance to separate resistance problems from stiffness problems.

$$C_{dyn} = \frac{V_T}{PIP - PEEP_{total}}$$

V_T tidal volume (mL)

$PEEP_{total}$ total PEEP (cmH_2O)

PIP peak inspiratory pressure (cmH_2O)

NORMAL / TARGET	50–80 mL/cmH_2O
ALERT	Large gap between Cdyn and Cstat (PIP much higher than P_{plat}): suggests elevated airway resistance
USED BY · LEVEL	Respiratory Therapist · Core

WORKED EXAMPLE

Inputs V_T 420 mL; PIP 35 cmH_2O; PEEP 8 cmH_2O

Step 1 35 − 8 = 27 cmH_2O

Step 2 420 ÷ 27 = 15.6 mL/cmH_2O

Cdyn = 15.6 mL/cmH_2O

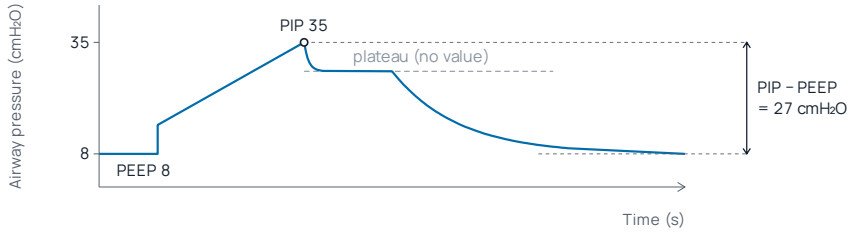


Figure 3.2 Dynamic compliance uses the peak inspiratory pressure, so it includes the pressure needed to overcome airway resistance as well as elastic recoil.

INTERPRETATION	Compare Cdyn with Cstat. If PIP − P_{plat} is large, suspect bronchospasm, secretions or a kinked or obstructed endotracheal tube (ETT).
COMMON MISTAKE	Interpreting Cdyn alone, without comparing it with Cstat, fails to separate resistive from elastic problems.
LIMITATION	Depends on inspiratory flow rate and pattern, so it is not a pure measure of lung elasticity as Cstat is.

SCAN SOURCE

Source Marino PL. Marino's The ICU Book. 4th ed. Philadelphia: Wolters Kluwer/Lippincott Williams & Wilkins; 2014.

See also Static compliance **p. 63** · Airway resistance **p. 67**

NUR-008 · CORE FORMULA

Norepinephrine Infusion

05

First-line vasopressor for septic and most distributive or vasodilatory shock per the Surviving Sepsis Campaign (SSC).

$$\text{Rate}\,(\text{mL/h}) = \frac{\text{Dose}\,(\text{mcg/kg/min}) \times W\,(\text{kg}) \times 60}{C\,(\text{mcg/mL})}$$

Dose	ordered dose (mcg/kg/min)	W	weight (kg)
C	concentration of prepared infusion (mcg/mL)		

NORMAL / TARGET	**Starting dose 0.01–0.05 mcg/kg/min, titrated to the MAP goal. FDA label (norepinephrine bitartrate injection, adults): initial 8–12 mcg/min, adjusted to desired effect; average maintenance 2–4 mcg/min (fixed-rate labeling, not weight-based).**
ALERT	**SSC 2021 remark: vasopressin is usually started when the norepinephrine dose is 0.25–0.5 mcg/kg/min; the guideline suggests adding vasopressin rather than escalating norepinephrine alone.**
USED BY · LEVEL	**Nurse · Physician · Pharmacist · Core**

WORKED EXAMPLE

Inputs: Dose 0.1 mcg/kg/min; weight 70 kg; concentration 16 mcg/mL

Step: (0.1 × 70 × 60) ÷ 16 = 26.25 mL/h

26.25 mL/h

Set the pump to 26.25 mL/h and titrate to the MAP goal

INTERPRETATION	Titrate to a MAP goal of ≥ 65 mmHg (or an individualized target) and reassess frequently.
COMMON MISTAKE	Giving norepinephrine through a peripheral line for an extended period without extravasation precautions. A central line is preferred.
LIMITATION	Peripheral extravasation carries a risk of tissue necrosis. The dosing weight differs between institutions; follow the local protocol.

SCAN PubMed

Source Evans L, Rhodes A, Alhazzani W, et al. Crit Care Med 2021;49(11):e1063-e1143

See also mcg/kg/min to mL/h **p. 135**

NUR-009 · CORE FORMULA

Epinephrine Infusion

05

Calculates epinephrine infusion rates for second-line vasopressor use in septic shock, infusion for refractory anaphylactic shock and inotropic or vasopressor support in cardiogenic shock.

$$\text{Rate (mL/h)} = \frac{\text{Dose (mcg/kg/min)} \times W\ \text{(kg)} \times 60}{C\ \text{(mcg/mL)}}$$

Dose ordered dose (mcg/kg/min)

W weight (kg)

C concentration of prepared infusion (mcg/mL)

NORMAL / TARGET	**Starting dose 0.01–0.05 mcg/kg/min. FDA label (epinephrine in sodium chloride premix, septic shock hypotension): titrate 0.05 to 2 mcg/kg/min to the desired blood pressure; wean gradually.**
ALERT	**No validated upper dose; the FDA label range runs to 2 mcg/kg/min. Monitor rhythm and lactate at every dose.**
USED BY · LEVEL	**Nurse · Physician · Pharmacist · Core**

WORKED EXAMPLE

Inputs	Dose 0.05 mcg/kg/min; weight 60 kg; concentration 16 mcg/mL	11.25 mL/h
Step	(0.05 × 60 × 60) ÷ 16 = 11.25 mL/h	Set the pump to 11.25 mL/h and titrate to the desired blood pressure

INTERPRETATION	Monitor for epinephrine-induced lactic acidosis, which can falsely suggest worsening shock.
COMMON MISTAKE	Confusing infusion dosing (mcg/kg/min) with cardiac arrest bolus dosing (1 mg IV push) is an error of scale.
LIMITATION	Epinephrine causes marked tachycardia and increases myocardial oxygen demand. Use caution in ischemic heart disease.

SCAN PubMed

Source Evans L, Rhodes A, Alhazzani W, et al. Crit Care Med 2021;49(11):e1063-e1143

See also Norepinephrine infusion **p. 138** · lactate clearance **p. 281**

ADULT CRITICAL CARE

The numbers behind adult critical care, worked through at the bedside.

The ICU Equations brings together 300 formulas, calculations and scoring systems used in adult intensive care, from blood gases and ventilator mechanics to vasoactive dosing, renal replacement therapy and ECMO.

Each entry sets out the equation with its variables and units, normal and alert values, a worked example, and the common mistakes and limitations that matter in practice.

300
entries in 16 parts, in clinical teaching order

Step by step
every worked example calculated line by line

Sourced
primary references, linked by QR code on every entry

23
waveforms and curves drawn from the worked examples

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12 Trauma & emergency
13 Neurocritical care
14 Research & statistics
15 Cross-specialty scores
16 Supplementary

For intensivists, respiratory therapists, critical care nurses, pharmacists, dietitians and physiotherapists.



ABOUT THE AUTHOR

Mohsin Almanami comes from a background in respiratory care and works alongside critical care teams in Bahrain in the field of medical technology. His focus is the quantitative side of intensive care: ventilation, oxygenation, hemodynamics and the bedside calculations that guide them.